Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- **Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors** — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors

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Step 1: Eigenvectors and Eigenvalues of $A^T A$

We start by looking at the matrix σ_j^2. We know that eigenvectors of $A^T A$, denoted v_j, correspond to the directions in which the matrix A stretches or compresses data, and σ_j^2 are the eigenvalues associated with these eigenvectors, which represent the amount of stretching.

The key equation here is:

$$A^T A v_j = \sigma_j^2 v_j$$

What does this mean? It tells us v_j is an eigenvector of the matrix $A^T A$, and σ_j^2 are the squared singular values of A. This is a critical step because it reveals the "directions" (the v_j's) in which A operates and how strongly it acts in those directions (through σ_j^2).

Step 2: Multiplying by A

To see how A transforms data, we multiply both sides of the equation by A:

$$A\left(A^T A v_j\right) = A\left(\sigma_j^2 v_j\right)$$

This simplifies to:

$$AA^T\left(A v_j\right) = \sigma_j^2\left(A v_j\right)$$

What does this tell us? It shows that Av_j is an eigenvector of AA^T. This is important because Av_j will help us construct the matrix U in the SVD.

Step 3: Normalizing to Find U_j

Now that we know Av_j is an eigenvector of AA^T, we normalize it to obtain u_j, which will become one of the columns of the matrix U in the SVD. The formula for u_j is:

$$u_j = \frac{A v_j}{\sigma_j}$$

This gives us a normalized vector u_j, which means its length is 1. So, u_j is the left singular vector, and it corresponds to the direction in which A stretches data.

Step 4: Constructing the SVD

Once we have all the singular vectors u_j and v_j and the singular values σ_j^2, we can construct the full SVD of A. The SVD is a factorization of A into three matrices:

$$A = U\Sigma V^T$$

Where:

1. U is the matrix of left singular vectors (from AA^T).
2. Σ is a diagonal matrix with the singular values σ_j^2.
3. V^T is the transpose of the matrix of right singular vectors (from $A^T A$).

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